

Visionary

13+3 · Local WM module · TENTATIVE

Fast stable teachable JEPA-WM brick. Pair with 2-AC (robot) + $\pi_{0.5}$ (motor).
BEHAVIOR: local module beside VLA — not whole stack.

NO Discord / Drive content in this deck

SPINE: Stable E2E JEPA WM · MSE+SIGReg only · $\sim 15\text{M}$ · $48\times$ vs DINO-WM (0.98s/47s) · SIBLING to Meta JEPA · SIM-ONLY · PushT 96 ± 2.83 · fixed-FLOP 90vs13 · recon hurts · λ -only

BEHAVIOR honesty

What wins today vs what LeWM offers

BEHAVIOR'25/'26 winners (typical)

$\pi_{0.5}$ + flow-matching action experts + SigLIP/PaliGemma — generative motor policies, not tiny E2E JEPA WMs.

LeWM shared interface

Image-goal CEM in latent space — same planning interface V-JEPA 2-AC popularized. Attractive as a local world-model module beside a VLA for real-time latent MPC — TENTATIVE deployment, not in-paper scores.

Do NOT invent LeWM robot/BEHAVIOR success rates — paper is Push-T / Cube / TwoRoom / Reacher only.

Local WM module — TENTATIVE

Beside the VLA · not the whole stack

Role

Fast latent imagination / image-goal subgoal scoring / surprise monitor

Not role

Whole BEHAVIOR policy · language goals · contact-rich motor expert

Why LeWM-shaped

Training stability + sub-second CEM on one GPU — edge-friendly brick

Label

TENTATIVE research proposal — zero in-paper BEHAVIOR numbers

Hybrid proposals — TENTATIVE

Research sketches · not paper claims

A. LeWM imagination + $\pi_{0.5}$ / FM motor

$\pi_{0.5}$ /FM executes short-horizon motor chunks; LeWM-style latent CEM scores long-horizon feasibility / image-goal subgoals / VoE residual as anomaly.

B. V-JEPA 2.1 state + LeWM-sized SIGReg predictor

Meta representation quality + LeWM latency: freeze/distill 2.1 features → compact AdaLN predictor with SIGReg for real-time MPC.

VoE as safety monitor — Fig. 8

Teleport surprise $\gg$ color · physical continuity alarm

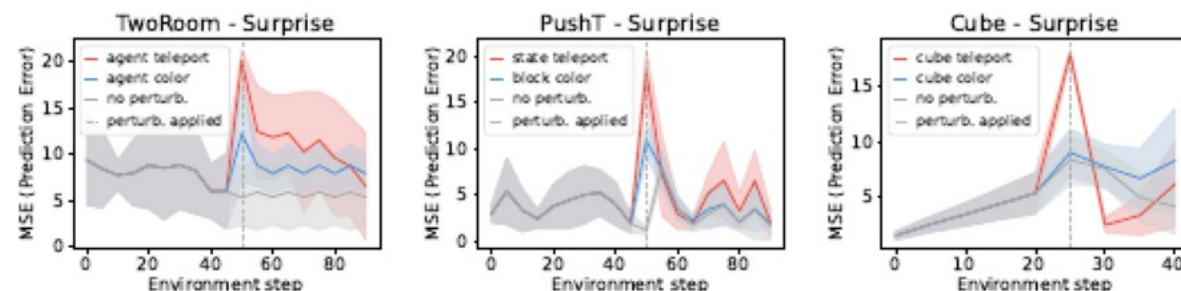


Figure 8: **Violation-of-expectation evaluation across three environments.** Each plot shows LeWM surprise along three trajectories: an unperturbed reference, a visually perturbed trajectory with abrupt object color change, and a physically perturbed trajectory where objects teleport to random positions. Teleportation violates physical continuity and produces a pronounced surprise spike, while the unperturbed trajectory stays at a low baseline. The increase is significant for teleportation across all environments (paired t-test, $p < 0.01$) but weaker and non-significant for color changes, indicating greater sensitivity to physical than visual perturbations. Environments, left to right: *TwoRoom*, *PushT*, *OGBench Cube*.

increasingly straight on PushT over training as a purely emergent phenomenon, without any explicit regularization encouraging this behavior, cf. Fig. 17. Remarkably, LeWM achieves higher temporal straightness than PLDM, despite PLDM employing a dedicated temporal smoothness regularization term. We detail our findings in App. H.

5.2 Violation-of-expectation Framework

Another approach to quantifying physical understanding is the ability to detect violations of the learned world model. Inspired by the violation-of-expectation (VoE) paradigm used in developmental psychology and recently adopted in machine learning [46–48], this framework evaluates whether a

TENTATIVE app: contact-rich slip / unexpected object motion alarms for teleop assist.

Hierarchical extension

Paper admits short $H=5$ — multi-room needs hierarchy

Limit today

Horizon $H=5$ ($\times$ frame-skip 5 $\rightarrow$ 25 env steps). Fine for Push-T; thin for household multi-room.

Paper future work

Hierarchical WM for long horizon; large video pretrain; inverse dynamics to drop action labels.

BEHAVIOR ask

Multi-room household tasks need subgoal imagination + motor chunks — hierarchy is the bridge.

Proposal

Coarse LeWM (or 2.1) subgoals $\rightarrow$ fine LeWM-sized CEM $\rightarrow \pi_{0.5}$ chunk execution.

Follow-up research agenda

Hypothesis-ready · no Discord/Drive

1. SIGReg-E2E on BEHAVIOR-1K / OXE play subsets — measure planning Hz vs DINO-WM.
2. Image-goal CEM inside BEHAVIOR scenes with LeWM latents vs language-goal VLA success.
3. VoE surprise as safety monitor for contact-rich skills.
4. Hierarchical LeWM for multi-room household tasks.
5. Distill V-JEPA 2.1 → 15M LeWM encoder for edge robots.
6. Adaptive SIGReg for low-intrinsic-dim envs (TwoRoom fix path).

New apps — beyond Discord/Drive

Practical surfaces only

Tabletop rearrange

Real-time latent MPC with image goals

Offline what-if

Policy eval inside compact JEPA imagination
(WorldGym-style)

Physics alarms

VoE residuals for teleop assist / safety

Teaching artifact

Smallest complete JEPA-WM you can train overnight

Edge MPC

15M brick on single GPU / Jetson-class

Subgoal scorer

Beside VLA — score imagined image-goal feasibility

Anti-patterns

Do not do these

Invent BEHAVIOR SR

Paper has none — label all robot/household claims TENTATIVE.

Replace 2-AC

Real-robot JEPA planning cite stays on V-JEPA 2-AC.

Whole-stack pitch

LeWM is a brick beside VLA/FM — not the BEHAVIOR answer alone.

Discord/Drive plans

Out of scope for this Visionary deck — skip.

Ignore TwoRoom

Low-dim failure mode is a research opportunity, not a footnote to hide.

Confuse recipes

E2E SIGReg $\neq$ freeze-then-AC $\neq$ $\pi_{0.5}$ FM motor.

**LeWM is the fast, stable, teachable
JEPA world-model brick.**

Pair it with V-JEPA 2-AC (real robot)
and $\pi_{0.5}$ (motor skill) —
not any one as the full BEHAVIOR answer.

Exam bite — Visionary

30 seconds · no Discord/Drive

Treat LeWM as a local latent-MPC / imagination module you can train overnight and run in under a second — TENTATIVE beside a VLA on BEHAVIOR, not instead of $\pi_{0.5}$ motor or V-JEPA 2-AC robot evidence. Hybrid: LeWM (or 2.1+SIGReg) for long-horizon image-goal scoring + VoE safety; FM/ $\pi_{0.5}$ for contact-rich chunks. Hierarchical for multi-room. Never invent LeWM household scores.

Numbers tattoo — pin these

Shared spine across all five seats

~15M

params · 1 GPU / few h

0.98s vs 47s

~48× planning speed

~200×

fewer tokens vs DINO

90 vs 13

fixed-FLOP Push-T

74 vs 48

fixed-FLOP Cube

96.0±2.83

Push-T SR (Tab.5)

78±5 / 92±1.63

PLDM / DINO Push-T

$\lambda \in [0.01, 0.2]$

SR>80%; peak ~0.09

96 → 86

recon loss hurts (T7)

SIBLING

not Meta descendant

SIM-ONLY

cite 2-AC for robots

λ -only

vs PLDM 6 knobs

Visionary — close

Takeaways + exam bite

1. BEHAVIOR: local WM module TENTATIVE beside VLA — not whole stack.
2. Hybrid: LeWM imagination + $\pi_{0.5}$ /FM motor; or 2.1 state + LeWM-sized SIGReg pred.
3. VoE safety monitor · hierarchical for multi-room · edge 15M brick.
4. One-liner: fast stable teachable JEPA-WM brick; pair with 2-AC + $\pi_{0.5}$. NO Discord/Drive.

EXAM BITE

Local brick, not whole stack. Pair LeWM with 2-AC (robot) and $\pi_{0.5}$ (motor). Label BEHAVIOR uses TENTATIVE.